



Department of Computer Science and Engineering

Academic Year 2021 – 2022 (Odd Semester)

Degree, Semester & Branch: III Semester B.E. ECE 'A'

Course Code & Title: EC8393 & Fundamentals of Data Structures in C

Name of the Faculty member (s): Ms.P.Jothi Thilaga, AP/CSE

Innovative Practice Description

- **Unit / Topic:** Unit IV / Trees & Tree Traversals
- **Course Outcome:** CO4
- **Topic Learning Outcome:** TLO10
- **Activity Chosen:** Jigsaw
- **Justification:**

Tree Traversal is an important concept in data structures. Different binary trees were assigned to students under each group. Each student in groups was given a chance to excel themselves in a particular topic and share that knowledge with their team members. The main theme of this Jigsaw activity is to promote both individual and peer teaching which requires students to understand the concept very clearly and engage in discussion and learning.

- **Time Allotted for the Activity:** 20 Minutes
- **Details of the Implementation:**

Eight batches of students group were formed based on the students' academic performance and ability. It is called homegroup. One student from each group was selected as the leader. Expert groups were formed that consisted of a student from home groups. 5 minutes were allotted to the teams to discuss the method of evaluation. The expert group students were brought back into their home groups and allotted 5 minutes. For further discussion was made, in which each member as an expert in the entire tree concepts – tree traversal, conversion of infix to postfix, and postfix evaluation. One student from each group has discussed the problem solutions with the other batch of students in the class. Others in the group were encouraged to raise queries for clarification. Problem solutions provided by the students are given as screenshots in Figure 1.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PSO1
CO1	3	3	2	2	2

(1 – Low 2 – Moderate 3 – High)

• PO / PSO mapped:

Innovative practice	PO1	PO2	PO3	PO4	PSO1
	3	3	2	2	2
Justification for correlation	Requires basic engineering knowledge to do tree traversals	Able to Identify engineering problems with tree structures concept	Able to design solutions for complex engineering problems using tree data Structures	Able to provide complex problem solutions with relevant tree traversal methods	Able to apply tree data structures to solve various application in Information & communication Industry

• Images / Screenshot of the practice:

I. Haaris
M. Badri Chelva
P. Gowtham
E. Jackson

Given the expression, Exp = $a + b/c * d - e$
construct the corresponding Binary tree.

$a + b/c * d - e$
 $[(a + (b/c) * d) - e]$

Given the Binary tree write down the expression

Prefix: left right root
 $a b / c d * + e f g h i j k l m n o$

Postfix: root left right
 $a + / a b * c d / * e f g - h i$

Infix: left root right
 $a / b + c * d * e f / g / h - i$

NAME : MAHA VIDHYA V
REG NO: 953620106045

B. Jayaramlona
M. Kirupa
G. Gokulpraga

Inorder Traversal : BDAEHGIFC
Preorder Traversal : ABDCEFGHIE
Postorder Traversal : DBHIGIFECA

Figure 1: Solutions presented by each group

- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

- All the students were actively participated and done the task and practice among their group members.
- Each student explored the knowledge of the tree concepts and tree traversal concepts among the other students in the class.
- It makes the students to gain well knowledge on tree traversal concept.

- ❖ ***Benefit of the practice:***

- Students were actively participated in this activity.
- From this activity, the students got clear idea about different tree traversals.
- Slow learners are also able to perform all the tree traversals.

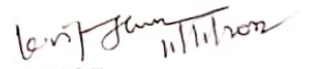
- ❖ ***Challenges faced in implementation:***

- Making students group was little difficult and takes time.
- Some students in the team did not participate actively in the team.

References:

- ❖ <https://www.jigsaw.org/>
- ❖ https://www.ritrjpm.ac.in/images/computer-science/8_OIT551_Jigsaw.pdf
- ❖ https://www.ritrjpm.ac.in/images/computer-science/36_IT6801_Jigsaw.pdf


Signature of Faculty Member


HOD